

Problem 1

1. Assuming R1 and R2 are union-compatible:
 - a. Min: N2, Max: N1 + N2
2. Assuming R1 and R2 are union-compatible:
 - a. Min: 0, Max: N1
3. Assuming R1 and R2 are union-compatible:
 - a. Min: 0, Max: N1
4. Min: N1*N2, Max: N1*N2
5. Assuming R1 has an attribute 'a'
 - a. Min: 0, Max: N1
6. Assuming R1 has an attribute 'a' and N1 > 0
 - a. Min: 1, Max: N1
7. Assuming the set of attributes of R2 is a subset of the attributes of R1
 - a. Min: 0, Max: 0

2.

A. How many copies of the book titled “ManchesterCityTrophies” are owned by the library branch whose name is “Liverpool”

$$\Pi_{NoOfCopies}(\sigma_{Title = 'ManchesterCityTrophies' \text{ AND } BranchName = 'Liverpool'}(Book \bowtie BookCopies \bowtie LibraryBranch))$$

B. For each library branch that has the book titled “ManchesterCityTrophies”, find the number of copies it has of that book

$$\Pi_{BranchName, NoOfCopies}(\sigma_{Title = 'ManchesterCityTrophies'}(Book \bowtie BookCopies \bowtie LibraryBranch))$$

C. Retrieve the names of borrowers who do not have any books checked out

$$\Pi_{Name}(Borrower - \Pi_{Name}(Borrower \bowtie BookLoans))$$

D. Find the names of borrowers who have checked out all books

$$\Pi_{Name}(\Pi_{CardNo, BookID}(BookLoans) \div \Pi_{BookID}(BookCopes)) \bowtie Borrower$$

3.

A. Find the names and course grades for students who take “OS” or “DBMS”

$$\Pi_{SName, Grade}(\sigma_{CName = 'OS' \text{ OR } CName = 'DBMS'}(Students \bowtie Courses \bowtie Enrollment))$$

B. Find the names of students who have taken all courses that the student with Sno = 10 has taken

$$\Pi_{SName, Sdept}(((Students \bowtie Enrollment) \div \Pi_{Cno}(\sigma_{Sno = 10}(Enrollment))))$$

C. Find the name and department of students who do not enroll in the course with Cno = 3

$$\Pi_{SName, Sdept}(Students - \Pi_{Sno, SName, Sdept}(\sigma_{Cno = 3}(Enrollment \bowtie Students)))$$